A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 07

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 Primary:** succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) **1 Secondary:** succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) **1** Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form **1** The stable community that develops under this repeated disturbance is a plagioclimax **1 Examples:** chalk grassland / hay meadows maintained by grazing or mowing – lowland heath by burning – coppiced woodland – Machair grassland – Broads grazing marshes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deflected succession** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost **1** Rainforest and coral reefs are climax communities so conservation aims to minimise human impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Extreme environments (deserts, polar) are dominated by abiotic factors – few species tolerate them – populations fluctuate and food webs are simple so changes in one food species have large effects **1** Stable environments allow many species to evolve inter-species relationships – populations are regulated by biotic factors – complex food webs give stability (rainforests, reefs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
02.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Tracking:** continuous data on movements, territory size, migration routes and habitat use without disturbance, over large areas and long periods – acoustic tags in seawater – geolocators for small birds – more representative than occasional sightings **2 DNA:** identifies individuals, gene pool size / inbreeding and regional populations (tracing ivory, timber, fish) – eDNA detects rare or hidden aquatic species (great crested newt, invasive carp) from shed cells without capture **2 Image recognition:** software identifies individuals from unique markings (tiger stripes, whale shark spots, fin damage) in camera-trap / drone / satellite images, giving population size, lifespan, social structure from large image sets quickly and objectively **2 Acoustic:** bat detectors, hydrophones and sonograms record species that are hard to see (bats, cetaceans, insects), giving presence, abundance and activity continuously, day and night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **validity** → actually measures what it is meant to measure • **gene pool** → the range of alleles / genes present in a population	8
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Data collected continuously / over large areas / at night rather than at a few times and places a researcher can visit **1** Less disturbance so animals behave naturally – larger sample sizes – objective rather than subjective identification **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-406; AQA spec 3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1 Diversity:** HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change **1** No single source could meet demand all year – battery storage on the island grid smooths fluctuations **1** Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
04.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife **1** Wood biomass heating – insulation such as straw-bale housing – some electric vehicles **1 Habitat protection and creation:** fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow) – eagles and hen harriers nesting **1** Recyclables shipped off – garden / food waste composted – low-impact food production – beach litter collection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Small groups can organise, agree and act quickly – decisions benefit people who see the results directly **1** Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Renewable energy is abundant but geographically and temporally variable – a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand **1** DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	
05.1	1 mark for the point below. **1** Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.2	**1 mark each** 1 mark for each of the 4 points below. **1** Age / date of the sample (radioisotope dating) **1** Past temperatures **1** Composition of the past atmosphere (eg CO_2 / methane concentrations) **1** **Temperature:** more magnesium is incorporated into calcite at higher temperatures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present **1** Ocean currents redistribute heat and nutrients – changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** There was no oxygen / ozone layer so high levels of UV radiation reached the land surface – water absorbed the UV and protected organisms in the oceans **1** Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely **1** The atmosphere contained toxic gases such as ammonia – the early Archaea were anaerobic so did not need oxygen **1** Water acted as the solvent for reactions / transport **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first 3 - the question asks for 3, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 600 + (90 + 150) - (220 + 80) = 540 **1** Stock fell by 60 so the catch exceeds the MSY / unsustainable **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
06.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** A + G - M = 90 + 150 - 80 = 160 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Catch is landed and weighed at ports so it is recorded **1** Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed **1** Representative sampling of a large enough proportion of the area is impractical / expensive – survival of eggs and larvae varies greatly between years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.4	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 CPUE:** if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained – a better measure than catch size which technology can prop up **2** Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out **1** Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth **1** The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding **1** Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients **1** Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances **1** Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving) – specialised flowers reduce pollen being wasted on other species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 3 points below. **1 Dispersal is not haphazard:** animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive **1** Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling **1** Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings **1** B is warmer / longer thermal growing season because south-facing slopes receive more insolation – valley floors are frost pockets where cold dense air collects **1** But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3
08.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) **1** Woven plastic cloches / fleece over rows to retain warm air near the ground **1** Burning oil / gas or using large fans in orchards to prevent frost damage to blossom **1** Site selection (aspect, avoiding frost pockets) – heated buildings and shelter for livestock **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) **1** Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Long days increase milk production and egg laying – poultry grow best with short days – sheep mate as days shorten in autumn **1** Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1000 / 100 000 x 100 = 1% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some light is reflected by the leaves **1** Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed **1** Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) **1** After four levels too little energy remains to support a further population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) **1** Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) **1** So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Protected areas:** Korup NP Cameroon (1260 km^2, 600 tree species, endemic Ancistrocladus korupensis with anti-HIV / antimalarial alkaloids) – Tumucumaque NP Brazil (39 000 km^2, adjoining French Guiana reserve) – Alto Maues – Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps – sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation – ITTO aims (largely unmet) – CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia) – community projects and employment – carbon sequestration payments • **Effectiveness:** protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict – drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest – climate change cannot be controlled locally • **Judgement:** combination of large protected areas, economic alternatives for local people and reduced international demand needed – overall loss continues **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **endemic** → found naturally only in that one place • **sustainable** → meeting present needs without stopping future generations meeting theirs • **CITES** → the treaty that controls international trade in endangered species	9
	Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Protected areas are the core of rainforest conservation. Korup National Park in Cameroon protects 1260 km^2 with about 600 tree species, including the endemic *Ancistrocladus korupensis* whose alkaloids have anti-HIV and antimalarial activity; Brazil's Tumucumaque National Park covers 39 000 km^2 and adjoins a reserve in French Guiana; the Rio Bravo area in Belize is funded by charities, companies, ecotourism and selective logging of high-value mahogany that does not damage the forest long term. Debt-for-nature swaps have cancelled Guatemalan and Brazilian debt in return for protection, CITES listing restricts trade in Honduras mahogany, the ITTO sets sustainable timber aims, ecotourism and community employment give local people an income from standing forest, and carbon sequestration payments are emerging.	
	Where they are enforced, protected areas work, but they face illegal logging, mining, poaching, road building and conflict, and ecotourism itself has damaged sites in Costa Rica and Malaysia. The deeper problem is that the drivers of deforestation - population growth and international demand for beef, soya, palm oil, timber and minerals - originate outside the forest, and climate change threatens even well-protected areas; the ITTO's aims are largely unmet and overall loss continues.	
	The most effective schemes are those that combine large protected areas with economic alternatives for local people, such as Rio Bravo, but they can only slow the loss unless demand in importing countries also falls.	
	Weaker answer - Level 2 (4 marks)	
	Rainforests are protected by national parks where logging is banned, for example in Brazil. Debt-for-nature swaps mean a country's debt is cancelled if it protects its forest. Ecotourism brings money from tourists so the forest is worth more standing than cut down. Sustainable logging only takes some trees. These efforts have saved some areas but deforestation is still happening because of illegal logging and farming, so they have only been partly effective.	
	What makes the difference	
	• Level 2 covers the main types of effort and a fair conclusion, but the question says 'describe' - Level 3 gives named examples with figures (Korup, Tumucumaque, Rio Bravo) as the mark scheme expects.	
	• Level 3 explains *why* effectiveness is limited (drivers originate outside the forest, ITTO aims unmet, climate change) rather than just noting continued loss.	
	• The judgement in Level 3 identifies what works best and what would be needed; Level 2 stops at 'partly effective'.	
10.2	1 mark for the point below. **1** Carbon sequestration reducing climate change – genetic resources / medicines – rainfall regulation – global biodiversity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Ecological features:** cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web – specific abiotic requirements (temperature, light, salinity, low turbidity) • **Importance:** quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • **Vulnerability:** narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching – slow growth so slow recovery – physical damage – overfishing and food-web effects (crown of thorns) – introduced species – dependence on mangroves and seagrass • **Conservation:** MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Sustainability definitions – population and per capita impact – attitudes of rural and urban societies • **Negative feedback:** cloud cover and albedo, CO_2 and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • **Tipping points:** rare under negative feedback – large events (Permian methane hydrate) – self-sustaining change from human activities • **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides) – links to climate change chapter – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

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